viewed as forming a digital signal at 6.4 MHz, with each sample represented by 16 bits. The signal is then decimated from 6.4 MHz to 100 kHz, accomplished by deleting every 63 out of 64 samples. In actual operation, much more goes on inside of this device than described by this simple discussion.

Delta-sigma converters can also be used for digital-to-analog conversion of voice and audio signals. The digital signal is retrieved from memory, and converted into a delta modulated stream of ones and zeros. As mentioned above, this single bit signal can easily be changed into the reconstructed analog signal with a simple low-pass analog filter. As with the antialias filter, usually only a single RC network is required. This is because the majority of the filtration is handled by the high-performance digital filters.

Delta-sigma ADC's have several quirks that limit their use to specific applications. For example, it is difficult to multiplex their inputs. When the input is switched from one signal to another, proper operation is not established until the digital filter can clear itself of data from the previous signal. Delta-sigma converters are also limited in another respect: you don't know exactly *when* each sample was taken. Each acquired sample is a composite of the one bit information taken over a segment of the input signal. This is not a problem for signals encoded in the frequency domain, such as audio, but it is a significant limitation for time domain encoded signals. To understand the shape of a signal's waveform, you often need to know the precise instant each sample was taken. Lastly, most of these devices are specifically designed for audio applications, and their performance specifications are quoted accordingly. For example, a 16 bit ADC used for voice signals does not necessarily mean that each sample has 16 bits of precision. Much more likely, the manufacturer is stating that *voice signals* can be digitized to 16 bits of *dynamic range*. Don't expect to get a full 16 bits of useful information from this device for general purpose data acquisition.

While these explanations and examples provide an introduction to single bit ADC and DAC, it must be emphasized that they are simplified descriptions of sophisticated DSP and integrated circuit technology. You wouldn't expect the manufacturer to tell their *competitors* all the internal workings of their chips, so don't expect them to tell *you*.